4.5 Solving Polynomial Equations

Name _____ Period ____ Date _____

Algebra II 4.5 · Student sheet

I can…

- solve polynomial equations and find the zeros of polynomial functions.
- use the Rational Root Theorem to find possible rational solutions.
- write a polynomial function from its zeros, including irrational conjugates.

Vocabulary · fill in the missing word

Repeated solution — a factor that _____ more than once, such as $(x - 3)^2$

Rational Root Theorem — every rational _____ of $a_n x^n + \ldots + a_0 = 0$ has the form $\pm(\text{factor of } a_0)/(\text{factor of } a_n)$

Irrational Conjugates Theorem — if $a + \sqrt{b}$ is a zero of a polynomial with rational _____, so is $a - \sqrt{b}$

Part 1 · Volume puzzle

A concrete block has a volume of 210 cubic meters and dimensions x, 2x − 3, and 2x − 4 meters.

x	3	4	5	6
x(2x − 3)(2x − 4)				

x = _____ Why must x be greater than 2? _____

Part 2 · Notes and examples (fill in each note, then work the example)

Example 1

Solve $2x^4 - 12x^3 + 18x^2 = 0$.

Example 2

Find the zeros of $f(x) = x^3 - 4x^2 - 9x + 36$. Check by graphing in Desmos.

Rational Root Theorem: Possible rational zeros = ±(factors of _____) / (factors of _____)

Example 3

Find all real zeros of $f(x) = x^3 - 8x^2 + 11x + 20$.

Irrational Conjugates Theorem: If $a + \sqrt{b}$ is a zero, then _____ is also a zero.

Example 4

Write a polynomial function f of least degree with leading coefficient 1 that has the zeros 2 and $3 + \sqrt{5}$.

Part 3 · Partner practice

Partner 1

Solve $x^3 + x^2 - 6x = 0$.

Partner 2

List the possible rational zeros of $f(x) = 2x^3 + x - 3$.